$$
\begin{bmatrix}
f^1(e_1) & f^2(e_1) & \cdots & f^k(e_1) \\
f^1(e_2) & f^2(e_2) & \cdots & f^k(e_2) \\
f^1(e_3) & f^2(e_3) & \cdots & f^k(e_3) \\
\vdots & \vdots & \ddots & \vdots \\
f^1(e_k) & f^2(e_k) & \cdots & f^k(e_k)
\end{bmatrix}
\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}
$$

$$
L(e_i) = \begin{bmatrix} f^i(e_1) \\ f^i(e_2) \\ \vdots \\ f^i(e_k) \end{bmatrix}
$$

L

$V_1(b)$ $\qquad$ $V_2(b)$

$\exists\, \lambda \in \mathbb{F}$ $\quad V_1(b) \cdot \lambda = V_2(b)$

$$V_1(\lambda b) = V_2(b)$$

$b_1 \qquad b_2$

$\lambda_1 \qquad \lambda_2$

$$\lambda_1 v_1(b_1)$$

$$\mathbin{\rotatebox{60}{=}}$$

$$v_1(\lambda_1 b_1) = v_2(b_1)$$

$$\lambda_2 v_1(b_2) = v_2(b_2)$$

$$\parallel$$

$$\lambda \cdot v_1(b_1)$$

$$v_1(b) \in \mathbb{F}^*$$

$$\bigcup_{\lambda \in \mathbb{F}} v_1(\lambda \cdot b) = \mathbb{F}^*$$

$$v_1(\lambda_2 b_2 - \lambda b_1) = 0$$

$$\lambda_2 b_2 - \lambda b_1 \in \ker v_1$$

$$v_2(\lambda_2 b_2 - \lambda b_1) = 0$$

$$= \lambda_2 v_2(b_2) = \lambda v_2(b_1)$$

$$= \lambda \cdot \lambda_1 v_1(b_1)$$

$$\frac{V_1(b_1)}{V_2(b_1)} \neq \frac{V_1(b_2)}{V_2(b_2)}$$

$$V_1 \left(V_2(b_2) \cdot b_1 - V_2(b_1) \cdot b_2 \right) \neq 0$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

V_1

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix}$$

$$(1-\lambda)(3-\lambda) + 1$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 \\ 4 & 4 \end{bmatrix} \quad 12$$

$a_1 = 1 \qquad a_{n+1} = 2^{n-1} + a_n \cdot 2$

$$a_{n+1} \cdot 2 = 2^{n+1} +$$

$$b_n = a_n - 2^n = a_{n-1} \cdot 2$$

$$b_2 = 2$$

$$b_n = 2^{n-1}$$

$$a_n = \underbrace{2^n + 2^{n-1}}$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \quad \begin{matrix} 4 & 0 \\ 4 & \end{matrix}$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2^n & 0 \\ a_n & 2^n \end{bmatrix} \quad \begin{matrix} 2^{n+1} & 0 \\ 2 \cdot a_n + 2^n & 2^{n+1} \end{matrix}$$

$$a_1 = 1 \qquad a_{n+1} = 2a_n + 2^n$$

$$n \cdot 2^{n-1}$$

$$(n+1) \cdot 2^n = n \cdot 2^{n-1} \cdot 2 + 2^n$$

$$b_{n+1} = a_{n+1} - 2^n = 2a_n =$$

$$b_n = 2a_{n-1}$$

$$b_2 = 2 \cdot a_1 = 2$$

$$b_2 = 2$$

$$\begin{matrix} 2 & -4 & 0 \\ 1 & -2 & 0 \\ 1 & -5 & 1 \end{matrix}$$

$$-(2-\lambda)(2+\lambda)(1-\lambda) + 4(1-\lambda)$$

$$(1-\lambda)\left(-4+\lambda^2+4\right)$$

$$\lambda^2(1-\lambda)$$

$$\overline{m}^{T}$$

$$\Downarrow$$

3

$$a_{n+1} = 2a_n + 2^n$$

$$a_{n+1} = 2\left(2 \cdot a_{n-1} + 2^{n-1}\right) + 2^n$$

$$n \cdot 2^n$$

$$x^{n+1}$$

$$nx^n \quad f\left(\begin{bmatrix} \lambda & & \\ 1 & \lambda & \\ & 1 & \lambda \end{bmatrix}\right)$$

$$\lambda$$

$$\begin{bmatrix} f(\lambda) & & \\ f'(\lambda) & f(\lambda) & \\ f''(\lambda) & f'(\lambda) & f(\lambda) \end{bmatrix}$$

$$M = S + S'$$

$$M^T = S^T + S'^T = S - S'$$

$$M + M^T = 2S$$